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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 12/21/2006 |
| **PI Name:** | Tanveer, Saleh Ph.D. | | |
| **Project Title:** | Dendritic Crystal Growth | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Dynamics and stability | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities |
| **Start Date:** | 01/01/2002 | **End Date:** | 07/18/2006 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 4 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 7/18/2006, per N. Hall (9/06) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Foster, Michael ( The Ohio State University ) | | |
| **Grant/Contract No.:** | NAG3-2700 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | We propose to investigate dendritic crystal growth in two and three dimensions, particularly when surface energy is small.<br><br>The particular goals are: (i) to study steady two and three dimensional dendritic crystal growth with different types of small effects, (ii) to recalculate the selection parameter for a non-axi-symmetric 3D dendrite that accounts for non-axi-symmetric corrections to Ivantsov paraboloid in a consistent manner, (iii) to determine if a singular perturbation of an otherwise smoothly evolving interface obtained before for one-sided 2-D model hold more generally for 3-D axisymmetric case and/or for two sided models.<br><br>Importance of the study is in predicting operating conditions for a dendrite, and determining if, given the near-structural instability for small-surface-energy anisotropy, there is a cross-over in the dependence of the selection parameter, usually denoted in the literature by ``sigma^* ''. Such a cross-over may serve to explain the absence of experimental evidence of a strong dependence on surface energy anisotropy as it tends to zero, as predicted by microscopic solvability. The time-evolution scenario for singular perturbations that we seek to study will help confirm the dynamical selection mechanism already found by the PI's for a two-dimensional (2D) one-sided model. In particular, it will solidify evidence against the marginal stability hypothesis that is still widely popular. Further, our coarsening result, if confirmed for 3D could set the stage for appropriate experiment to determine if, in the limit of small sigma^*, there is an intermediate range of distances from the dendrite tip where a power law different from a third (predicted in the far-field by Voorhees & Glicksman) is valid. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The impact of research is two fold.<br>There is a lot of contradictory claims made in the literature about selection in dendritic crystal growth. The mathematically rigorous part of our work includes precise theorems clarify these results considerably. More details are available in the progress report.<br><br>Work on the Bridgman crystal growth device by the Co-PI and collaborators can have potential impact on this industrial process of growing crystals. |
| **Task Progress:** | The progress made can be broadly classified into the following categories:<br>a) Tip Selection for 3-D Dendrites: Here, we have results for dendrites that are far from being paraboloid, as assumed in previous literature. We have significantly different results from the most widely known results of Brener. The results are being carefully prepared for an important publication.<br><br>b). Mathematically rigorous work on selection. In work with students and ex-students, rigorous work has been completed on exponential asymptotics of integro-differential equations that arise in the Saffman-Taylor finger as well as viscous fingering. This is the first set of work in this area and firmly establishes selection theory. While most researches usually accept this theory, they have been challenged in prestigious physics journals.<br><br>c) Effect of a distant interface. It was claimed recently by a well-known researcher that the distant effect of a second interface has the effect of selection of specific pattern, even without surface energy effects. In work with a collaborator, we showed otherwise through exact solutions.<br><br>d) Nonlocal evolution of localized disturbances in 2-D. A student and the PIs have considered the problem of small localized disturbance and obtained numerical results on their evolution. The nonlocal parameter free evolution equation sheds light on how nonlinear effects enter into previous linear calculations.<br><br>e) The co-PI and collaborators have considered the effect of rapid rotation on Bridgman crystal growth. No optimization of rotation rates in reducing segregation was found, as claimed by other investigators. |
| **Bibliography Type:** | Description: (Last Updated: 03/20/2017) |
| **Articles in Other Journals or Periodicals** | Liu X, Tanveer S. "Rigorous error bounds for computed Saffman-Taylor finger widths." Submitted October 2006. http://www.math.ohio-state.edu/vigre/pdf/report-activity-58.pdf , Oct-2006 |
| **Articles in Peer-reviewed Journals** | Xie X, Tanveer S. "Rigorous results in steady finger selection in viscous fingering." Archive for Rational Mechanics and Analysis, 2003 Feb;166(3):219-86. http://dx.doi.org/10.1007/s00205-002-0235-4 , Feb-2003 |
| **Articles in Peer-reviewed Journals** | Tanveer S, Xie X. "Analyticity and nonexistence of classical steady Hele-Shaw fingers." Communications on Pure and Applied Mathematics. 2003 Mar;56(3):353-402. http://dx.doi.org/10.1002/cpa.3030 , Mar-2003 |
| **Articles in Peer-reviewed Journals** | Crowdy DG, Tanveer S. "The effect of finiteness in the Saffman-Taylor viscous fingering problem." J Stat Phys. 2004 Mar;114(5-6):1501-36. http://dx.doi.org/10.1023/B:JOSS.0000013962.78542.33 , Mar-2004 |
| **Articles in Peer-reviewed Journals** | Xie X. "Rigorous results in selection of steady needle crystals." Journal of Differential Equations. 2004 Mar 1;197(2):349-426. http://dx.doi.org/10.1016/S0022-0396(03)00184-0 , Mar-2004 |
| **Articles in Peer-reviewed Journals** | Shyam V, Foster MR. "The effect of rapid rotation on a vertical Bridgman furnace at large Rayleigh number." Quarterly Journal of Mechanics and Applied Mathematics, 2007 May;60(2):201-30. http://dx.doi.org/10.1093/qjmam/hbm004 , May-2007 |
| **Dissertations and Theses** | Liu X. "Rigorous exponential asymptotics for a nonlinear third order difference equation." Dissertation, Ohio State University, December 2004. , Dec-2004 |